

Energy-system interactions reshape offshore hydrogen trade and carbon mitigation in Europe

Sheng Wang^{1,2} and Muhammad Maladoh Bah^{2,3}

^{1*}School of Engineering, Newcastle University, Newcastle University, Newcastle upon Tyne, NE17RU, The United Kingdom.

²School of Electrical and Electronic Engineering, University College Dublin, Belfield, Dublin, D04 V1W8, Ireland.

³Sustainable Energy Authority Of Ireland, 3 Park Place, Hatch Street, Dublin, D02 FX65, Ireland.

Contributing authors: sheng.wang@newcastle.ac.uk;
muhammad.maladohbah@seai.ie;

Abstract

Offshore wind-rich island systems are often assessed through production costs, but their hydrogen export value also depends on domestic absorption, shipping, demand geography and competing imports. Here we present the first integrated European-scale modelling framework linking offshore hydrogen cost-supply curves, hourly power-gas operation, cross-border hydrogen/ammonia shipping and European demand from 2030 to 2050. A lowest-LCOH-only allocation identifies Denmark or the Netherlands as leading exporters, whereas the integrated model identifies the UK as the leading net exporter in all three model years, with Ireland nearly matching it by 2050. In the integrated 2050 case, the UK and Ireland each reach approximately 162 TWh yr⁻¹ of net exports, and the full supply mix mitigates about 175.5 Mt CO₂ yr⁻¹ in the modelled displacement accounting, of which 158.3 Mt CO₂ yr⁻¹ is attributed to European offshore suppliers. The result is conditional on the external import option: delivered outside prices near 2–3 EUR kg⁻¹ H₂-equivalent displace most European offshore supply, whereas prices around and above 4 EUR kg⁻¹ restore the UK/Ireland export pattern. These findings show that strategic offshore hydrogen exporters cannot be inferred from LCOH alone.

Keywords: hydrogen, offshore wind, carbon mitigation, international trade

1 Introduction

Europe’s offshore wind expansion creates a spatial allocation problem, not only a generation-cost problem. Ireland and the UK have announced offshore wind development plans reaching 37 and 125 GW of installed capacity, respectively, by 2050 [1, 2]. These targets are more than twice the projected values of their total power demands in 2050 [3, 4]. Connecting and using such large offshore generation volumes through domestic power systems alone is challenging because domestic demand is limited and real-time balancing requirements increase with uncertain and intermittent offshore wind.

Green hydrogen offers a route for moving offshore renewable energy across sectors and borders. Renewable electrolysis avoids the direct operational emissions associated with grey hydrogen and can reduce lifecycle emissions relative to fossil-based production routes [5]. Hydrogen can also be stored and transported at larger scales than electricity, allowing offshore wind output to serve domestic industry, power and heat demands or distant demand centres through pipelines and shipping. This matters because prospective exporters such as Ireland and the UK sit near large future import needs: Germany expects annual imported hydrogen needs of 45–77 TWh by 2030 [6, 7], and Belgium has identified 20 TWh of imported hydrogen demand by 2030 [8].

Production cost remains central to offshore hydrogen deployment, but it is not sufficient for identifying strategic exporters. The levelised cost of hydrogen (LCOH) in 2030 has been projected at around 54–184 €/MWh, overlapping with blue hydrogen with carbon capture and storage and green hydrogen from onshore wind [9]. Country- and region-specific LCOH studies have assessed North Sea countries [10], the UK [11], Poland [12] and the Iberian Peninsula [13, 14], often with detailed cost breakdowns, production routes and uncertainty analysis. However, separate national studies cannot show how production costs interact with domestic absorption, shipping costs, European demand geography and competing imports.

This gap is particularly important for Ireland and the UK. European hydrogen strategies often emphasise imports from North Sea countries, North Africa, the Middle East and the Americas, while the export potential of these Atlantic-facing island systems is less quantified. Policy and consulting reports have identified opportunities for Irish and UK hydrogen exports [1, 15], but a pipeline-only or LCOH-only view can miss how their offshore resource value changes once domestic energy-system constraints and European trade balances are considered together. The missing element is not another standalone production-cost estimate; it is a coupled assessment of where offshore hydrogen can be produced, how much can be absorbed domestically, how surplus can move through hydrogen and ammonia trade, and how competing imports affect the resulting carbon-mitigation value (Supplementary Table 1).

This work provides the first integrated European-scale assessment of offshore hydrogen trade that links geospatial production costs, hourly domestic power-gas absorption, shipping-based hydrogen and ammonia exchange, competing external imports, an LCOH-only counterfactual and supplier-specific carbon-mitigation attribution in one optimisation framework. We first construct geospatial cost-supply curves for offshore hydrogen in 11 coastal countries using meteorological, bathymetric, infrastructure and exclusion-area data. We then use an hourly coordinated

power-gas optimisation model for Ireland to estimate how domestic energy systems absorb offshore wind and hydrogen under evolving operating constraints, and extend the resulting export-potential logic to the wider country set. Finally, we combine cost-supply curves, domestic export limits, shipping costs, national hydrogen and ammonia demand, and an external import option in an international trade optimisation model. This structure lets us test whether strategic exporters can be inferred from LCOH alone and how robust the resulting trade geography is to competing imported hydrogen.

2 Results

2.1 Offshore hydrogen costs do not determine export value alone

The first question is whether high-quality offshore wind resources also define the lowest-cost and therefore most strategic hydrogen exporters. They do not. We assess Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom (UK), which combine large exclusive economic zones (EEZs), offshore infrastructure potential and navigable links to European demand centres. Wind velocity, air pressure and temperature data are obtained from ERA5 at 1 hour and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions [16]. Hourly capacity factors and wake-effect losses are simulated using a 15 MW reference wind turbine [17]. Ireland has the highest modal wind speed among the studied EEZs, followed by the UK, because both are exposed to Atlantic wind resources (Fig. 1b).

Site-level costs change this resource ranking. We combine simulated capacity factors with development, capital, operating and decommissioning costs for offshore hydrogen production platforms, including turbines, foundations, electrolyzers and auxiliary systems (Methods). Bathymetry, port locations and power/gas connection points are included so that water depth, installation distance and maintenance distance affect site-level costs [18, 19]. The resulting LCOH ranges from 84.93 to more than 200 €/MWh across the studied EEZs (Fig. 1a). Denmark has the lowest-cost locations because it combines high capacity factors with a shallow continental shelf, although the lower-bound offshore hydrogen cost remains 33.94% higher than an onshore blue-hydrogen benchmark of 68.92 €/MWh [20]. The UK’s LCOH ranges from 85.97 to 307.26 €/MWh, while Ireland’s ranges from 90.42 to 201.8 €/MWh. Thus, Atlantic wind-resource quality does not automatically translate into the lowest production cost once foundation depth and infrastructure distances are included.

Cost-supply curves show how this geography evolves with deployment scale. We further account for wake losses, vessel density, excluded areas such as military and aquaculture zones, and future cost reductions [21] to construct country-level cost-supply curves for 2030, 2040 and 2050 (Fig. 2). In 2030, marginal LCOH is 76.10 €/MWh in Ireland and 79.02 €/MWh in the UK. Ireland is slightly cheaper at the policy-constrained marginal capacity, but the UK retains more low-cost near-shore supply. Denmark remains the leading cost benchmark, while France and the Netherlands retain competitive supply segments even when their full offshore capacity is not the lowest-cost option.

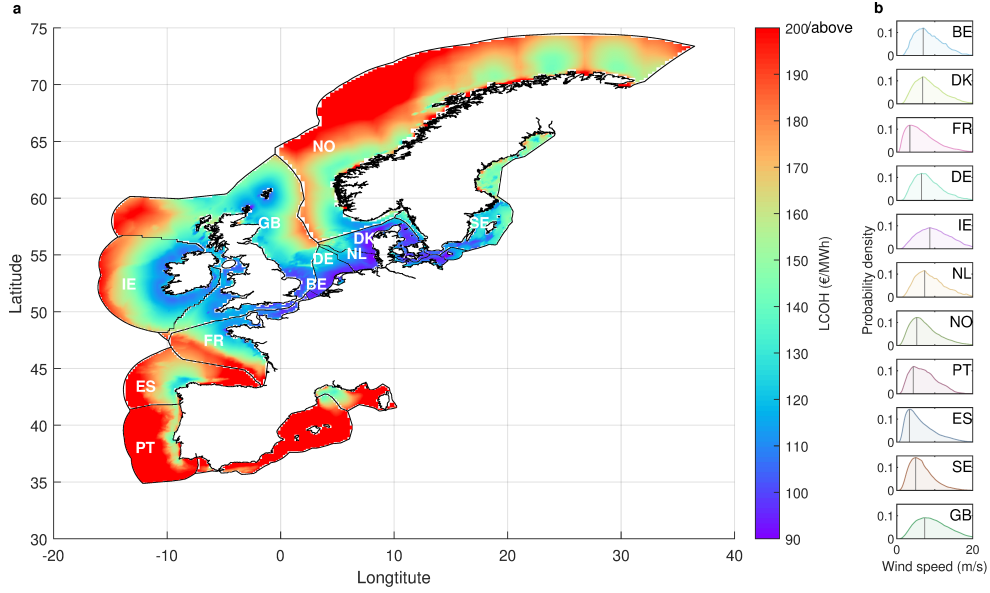


Fig. 1 Atlantic wind quality does not by itself determine the lowest offshore hydrogen cost. **a** LCOH in the EEZs of the studied coastal European countries, with black lines outlining EEZ boundaries. **b** Country-level wind-speed probability densities. The vertical grey line in each density plot marks the modal wind speed. BE, DK, FR, DE, IE, NL, NO, PT, ES, SE and GB denote Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom, respectively.

Technology and supply-chain learning reduce LCOH in later scenarios, but they do not make the Atlantic island systems simple lowest-cost winners. By 2050, marginal LCOH falls to 54.42 €/MWh for Ireland and 52.51 €/MWh for the UK, and all studied countries have offshore hydrogen segments that can compete with the assumed blue-hydrogen cost range (Fig. 3). The key implication is that Ireland and the UK have sufficient cost competitiveness to participate in offshore hydrogen trade, but their strategic value must be tested against domestic absorption, shipping and demand.

2.2 Domestic power-gas constraints create exportable offshore surplus

Export value appears only after domestic absorption begins to bind. Because the offshore wind ambitions of Ireland and the UK exceed their projected domestic power demands, we estimate how much offshore production can be absorbed domestically before it becomes available for export. We use a whole-energy-system optimisation model that simulates hourly power and gas operation over a full year under uncertain offshore wind generation. Power and natural gas systems are coupled through gas-fired power plants and onshore/offshore electrolyzers, allowing bidirectional energy conversion to expand system flexibility. Hydrogen blending and pure-hydrogen network cases are represented through different maximum hydrogen fractions in the gas

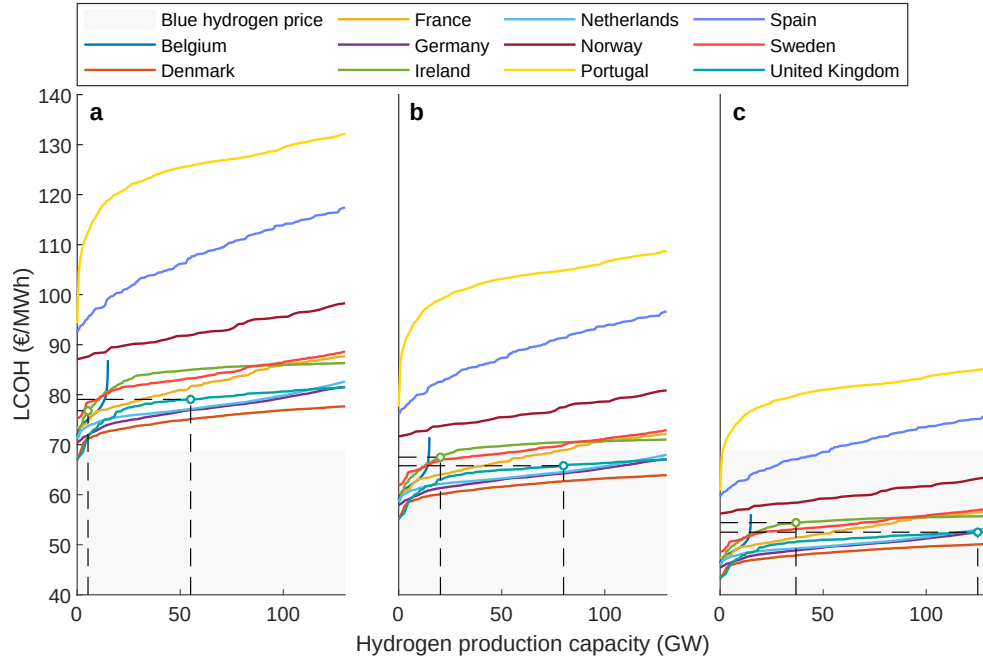


Fig. 2 Cost-supply curves show widening offshore hydrogen competitiveness but do not rank strategic exporters. a–c, Green-hydrogen cost-supply curves for 2030, 2040 and 2050, respectively. Solid lines indicate countries. Green and blue dots mark Ireland’s and the UK’s offshore wind capacity targets on their corresponding supply curves. Grey bands show the assumed future blue-hydrogen production-price range.

network. Hourly unit commitment includes the All-Ireland generator fleet, interconnectors, reserve, inertia, must-run and system non-synchronous penetration constraints, and observed hourly demand profiles (Methods; Supplementary Information) [22].

The domestic absorption result is strongly seasonal (Fig. 4). We assume a maximum 20% hydrogen blending ratio in 2040, reflecting near-term blending demonstrations and policy movement, and a pure-hydrogen network in 2050 in line with the UK’s hydrogen network roadmap [23]. Without export, the average summer wind-curtailment rate increases from 62.36% in 2030 to 69.80% in 2050 because high offshore generation coincides with low seasonal demand. In winter, the curtailment rate falls from 32.05% in 2030 to 13.47% in 2050 because higher hydrogen blending allows the gas system to absorb more offshore energy.

Even after extensive hydrogen use in the gas system, domestic demand cannot absorb the full offshore wind output. Export access substantially increases utilisation across all model years and seasons by creating an outlet for surplus hydrogen. In the 2050 summer case, for example, the average wind-curtailment rate falls to 10.46% with export access, and the full-year decomposition identifies approximately 170 TWh yr^{-1} of Irish export potential. Curtailment does not fall to zero because operating reserves, inertia requirements and must-run constraints still limit how much uncertain offshore output can be used at the margin.

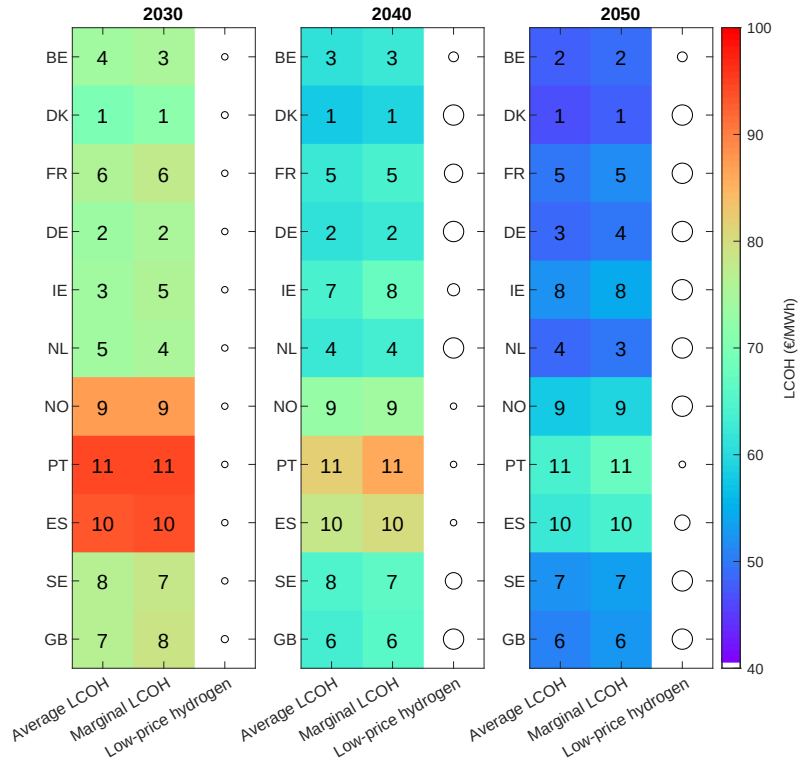


Fig. 3 Different cost metrics produce different country rankings for offshore green hydrogen. Rank indicates the average or marginal LCOH rank among the 11 studied countries. Marginal means the LCOH at the policy-constrained offshore capacity of that country in the corresponding year. Average means the capacity-weighted average LCOH across the offshore capacity in the corresponding year. Low-price hydrogen means the hydrogen production capacity with production cost below the upper bound of blue hydrogen. Circle size indicates low-price hydrogen capacity, with the smallest and largest circles representing 0 and >100 GW, respectively.

The same domestic-absorption logic changes the Europe-wide export screen. We use the full-year Ireland results to estimate offshore wind utilisation for the wider country set, assuming that domestic offshore absorption scales with electricity and gas demand. This is a screening-level extrapolation, not a substitute for country-specific unit-commitment studies; we therefore test its effect through a domestic-absorption sensitivity in the Supplementary Information. The annual decomposition identifies available export potential rather than optimised net trade (Fig. 5). In 2030, the UK has the largest available export potential at 176 TWh yr^{-1} because its offshore capacity is large relative to domestic load. The Netherlands, Denmark and Ireland follow with 64, 30 and 16 TWh yr^{-1} , respectively. In 2040, the UK and the Netherlands remain the two largest potential exporters, with 195 and 155 TWh yr^{-1} , while Ireland reaches 87 TWh yr^{-1} and becomes the third-largest potential exporter. By 2050, the UK and Ireland reach 263 and 170 TWh yr^{-1} , respectively, while the Netherlands' export potential falls because higher domestic hydrogen use absorbs more offshore output. Ireland therefore

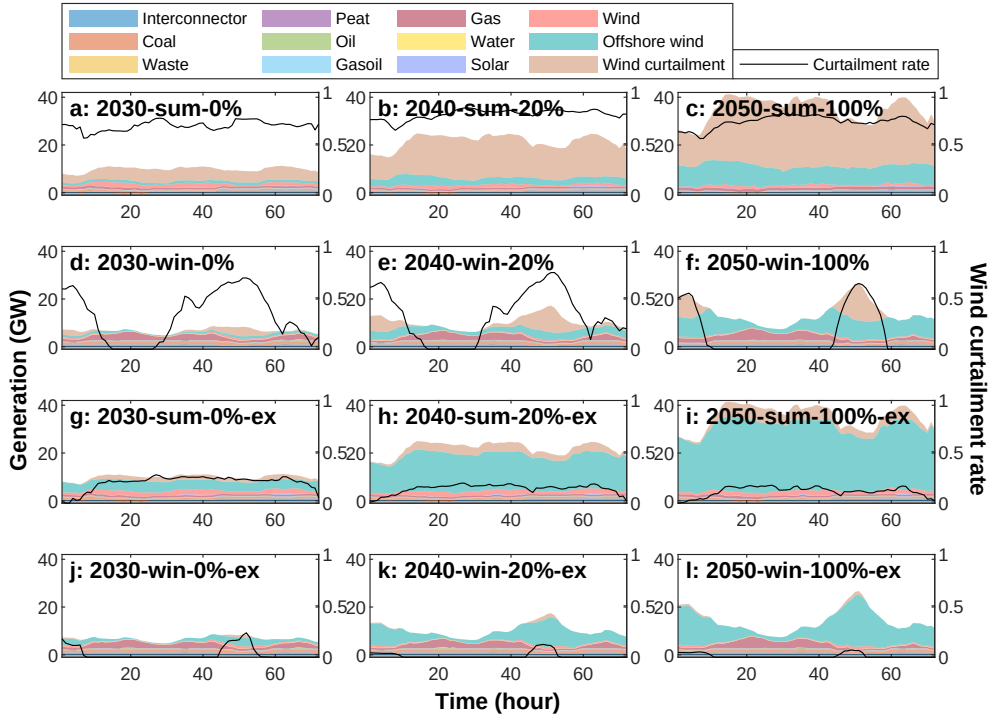


Fig. 4 Export access converts otherwise curtailed offshore wind into usable hydrogen output. Panels compare representative 72-hour summer (sum) and winter (win) periods in 2030, 2040 and 2050. The first two rows show cases without export access, and the last two rows show otherwise matched cases with export access (ex). Coloured areas show interconnector power and generation by technology, with offshore wind and curtailed offshore wind separated from other generation. Black lines show the offshore wind curtailment rate on the right axis. The 0%, 20% and 100% labels denote the allowed hydrogen share in the gas network, with 100% representing pure-hydrogen network operation.

becomes the second-largest source of available export potential, despite not being a lowest-LCOH country.

2.3 Integrated trade reallocates carbon-mitigation value across Europe

Available surplus becomes strategically important only if it can reach demand centres at competitive delivered cost. We therefore combine the cost-supply curves and export-potential estimates in an international green hydrogen trading model for 2030, 2040 and 2050. The cost-supply curves are interpolated as piecewise linear functions and combined with shipping costs, national hydrogen and ammonia demand, domestic export limits and external imports in a cost-minimising optimisation. The resulting flows and supplier contributions to carbon mitigation are shown in Fig. 6.

The optimised trade pattern is not a simple map of production cost. In 2030, the UK is the major hydrogen exporter, followed by the Netherlands, Denmark, and Ireland, with net exports of approximately 78, 39, 27 and 14 TWh yr⁻¹, respectively.

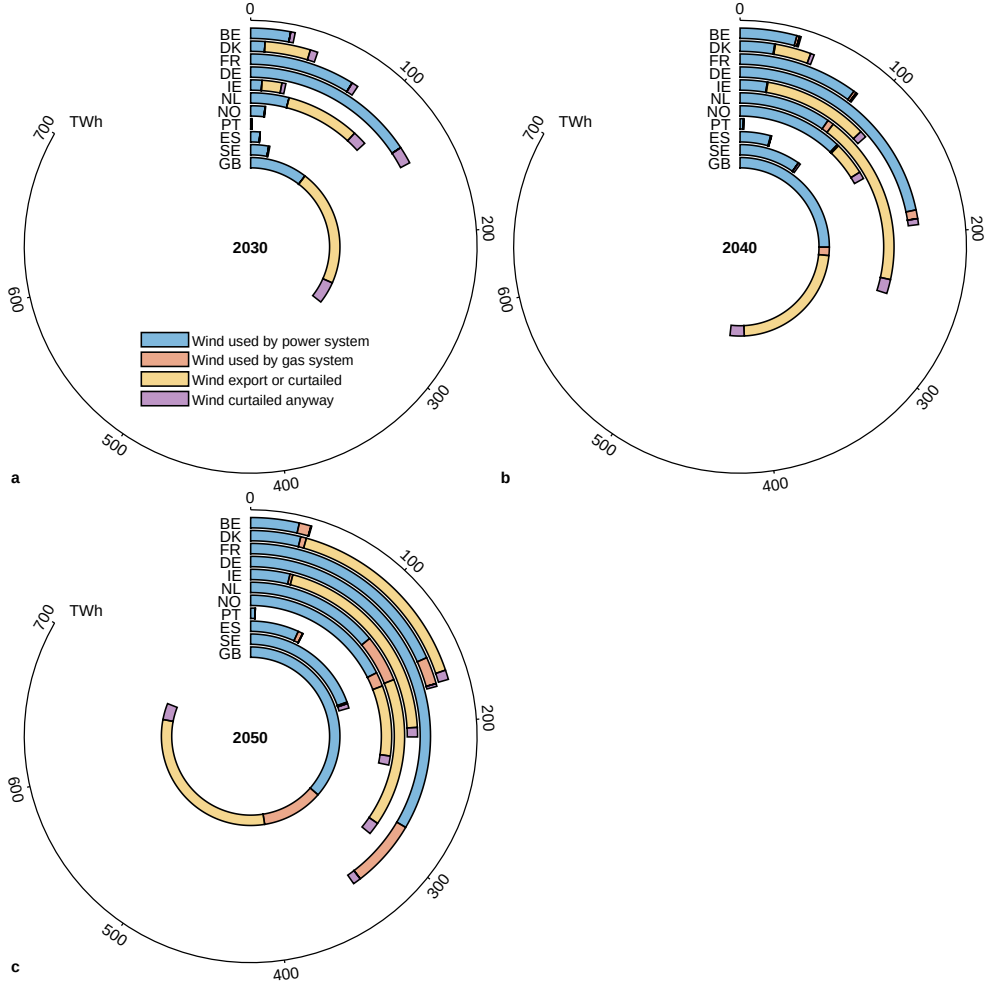


Fig. 5 Domestic power-gas absorption leaves uneven exportable surplus across Europe. **a–c**, Decomposition of annual offshore wind use across the 11 studied countries in 2030, 2040 and 2050. Each circular stack separates wind used by the power system, wind used by the gas system, wind available for export or otherwise curtailed, and residual wind curtailed anyway. Radial values are in TWh yr⁻¹. Country abbreviations follow Fig. 1. The UK remains the largest available-surplus system, while Ireland becomes the second-largest by 2050 despite not being a lowest-LCOH country.

Germany remains the major hydrogen receiver due to its high demand, so the lowest-cost and geographically close suppliers do not determine the whole trade pattern by themselves. By 2040, external hydrogen imports become necessary because offshore capacity growth is slower than demand growth. The integrated solution identifies approximately 116 TWh yr⁻¹ of external imports, while the UK, the Netherlands and Ireland remain large European net exporters with approximately 122, 85 and 83 TWh yr⁻¹, respectively. In 2050, external imports fall to approximately 62 TWh yr⁻¹ but remain part of the overall supply structure. The UK and Ireland become the two

largest European net exporters at approximately 162 TWh yr⁻¹ each, followed by Denmark at approximately 118 TWh yr⁻¹.

Hydrogen trade also reallocates carbon-mitigation value across Europe (Fig. 6b,d,f). In the modelled displacement accounting, the integrated supply mix mitigates approximately 43.5, 129.8 and 175.5 Mt CO₂ yr⁻¹ in 2030, 2040 and 2050, respectively, equivalent to 0.74, 2.20 and 2.98 times Ireland’s current annual carbon emissions [24]. Of these totals, European offshore suppliers account for approximately 43.5, 103.2 and 158.3 Mt CO₂ yr⁻¹, with the remaining contributions in 2040 and 2050 associated with external imports. In 2030, the UK and the Netherlands are the main European offshore contributors to carbon mitigation. From 2040 to 2050, Ireland’s importance increases with its export position. Ireland and the UK therefore mitigate emissions not only through domestic supply, but also through hydrogen delivered to demand centres elsewhere in Europe.

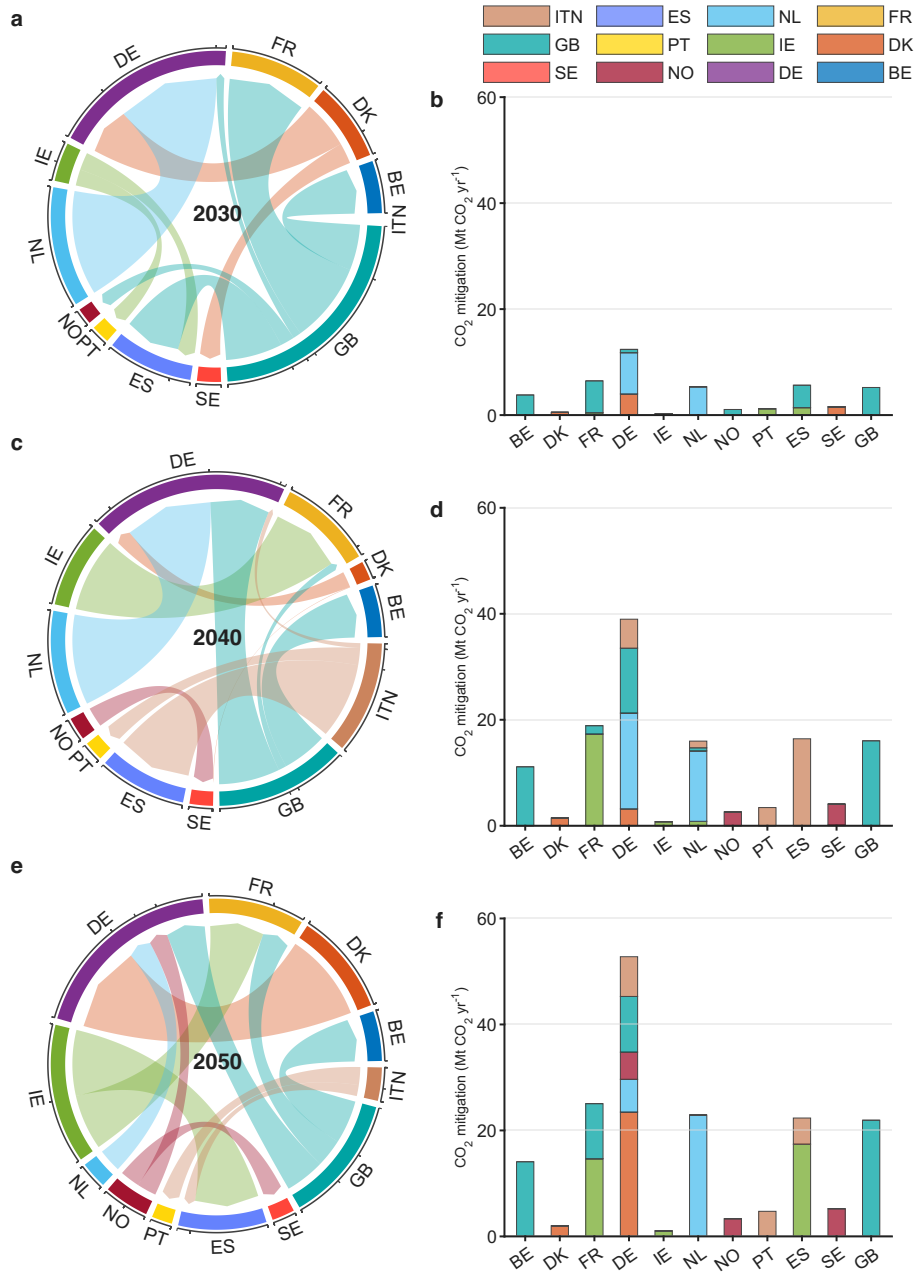


Fig. 6 Integrated hydrogen and ammonia trade reallocates carbon-mitigation value across Europe. **a, c** and **e**, Cross-border hydrogen-equivalent flows and external imports in 2030, 2040 and 2050, respectively. ITN denotes external imports; within-country self-supply is omitted from the flow panels. **b, d** and **f**, Supplier-attributed carbon mitigation, including domestic self-supply and exports. Bar colours identify the contributing supplier. ITN is included in plotted totals but excluded from European offshore supplier contributions reported in the text.

2.4 Production cost alone misidentifies strategic hydrogen exporters

The preceding results suggest a direct test of the paper’s central claim: would the same exporters be identified if demand were assigned only to the lowest-cost offshore supply? To isolate what is missed by a production-cost view, we constructed a lowest-LCOH-only counterfactual that allocates hydrogen demand to the lowest-cost offshore supply first, and compared it with the integrated model that includes domestic absorption limits, shipping costs, national demand and residual external imports. This counterfactual is not intended as a market-design proposal; it is a diagnostic baseline for testing whether the strategic-exporter ranking can be inferred from LCOH alone.

The two approaches give different exporter geographies (Fig. 7). In the cost-only baseline, the leading net exporters are Denmark in 2030, the Netherlands in 2040 and Denmark again in 2050, with net exports of 44, 147 and 171 TWh yr⁻¹, respectively. In the integrated model, the leading net exporter is the UK in all three model years, with 78, 122 and 162 TWh yr⁻¹, while Ireland nearly matches the UK in 2050 at 162 TWh yr⁻¹. The difference is particularly large in 2050: the cost-only baseline assigns 41 TWh yr⁻¹ of net exports to Ireland and a 44 TWh yr⁻¹ net import position to the UK, whereas the integrated model increases both countries to approximately 162 TWh yr⁻¹ of net exports. The integrated model also reveals residual external import requirements of approximately 116 and 62 TWh yr⁻¹ in 2040 and 2050, while the cost-only baseline hides this shortage by construction. These results show that the policy-relevant exporter ranking is determined by the system balance between cost, domestic absorption, shipping and demand, rather than by LCOH ranking alone.

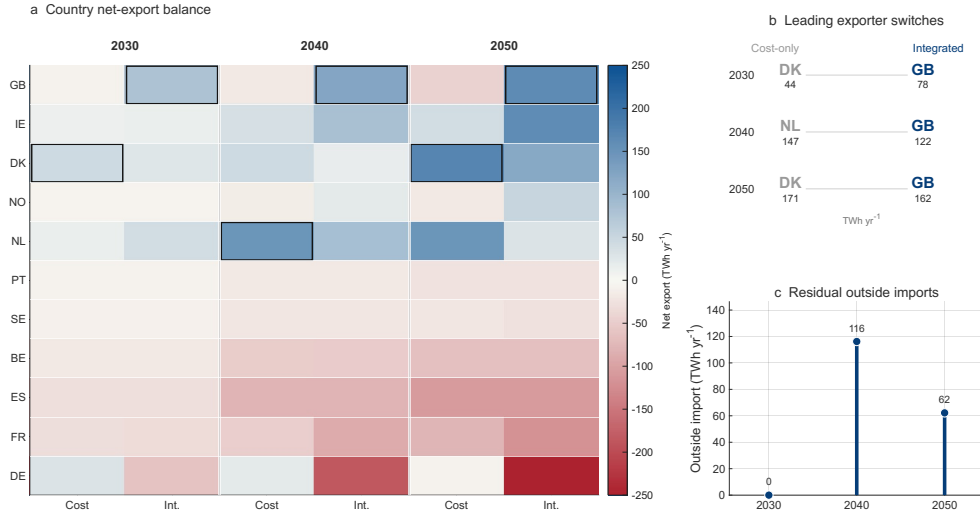


Fig. 7 A lowest-LCOH allocation misidentifies strategic offshore hydrogen exporters. **a** Country net-export balance under a lowest-LCOH-only baseline and the integrated model in 2030, 2040 and 2050. Blue indicates net export and red indicates net import. Black boxes mark the leading net exporter in each model-year combination. **b** Leading net exporter and corresponding net export under the two model structures. **c** Residual outside imports identified by the integrated model. The cost-only baseline allocates demand within the modelled European offshore supply by construction and therefore does not expose the residual external import requirement.

2.5 External import prices define a threshold for European offshore supply

The UK/Ireland export result is strong, but not unconditional. It depends on how attractive non-European low-carbon hydrogen or ammonia imports become relative to European offshore supply. This outside option is strategically important because European policy already anticipates substantial renewable hydrogen imports, while the future scale, timing and cost of global low-emission hydrogen trade remain uncertain [25–27]. We therefore replaced the previous high import penalty with a delivered hydrogen-equivalent outside-option price and solved the trade model over 2–6 EUR kg⁻¹ H₂-equivalent. This range should be read as a threshold scan rather than a price forecast: its low end is informed by optimistic global trade and cost estimates, while the upper range represents less favourable delivery, infrastructure, finance or policy conditions [28, 29].

The outside-option scan shows a threshold rather than unconditional robustness (Fig. 8). In the 2050 case, an external option near 2 EUR kg⁻¹ H₂-equivalent displaces European offshore supply almost entirely, with 818 TWh yr⁻¹ supplied from outside the modelled European offshore system. At 3 EUR kg⁻¹, European offshore supply remains limited to 98 TWh yr⁻¹, outside imports remain at approximately 721 TWh yr⁻¹ and the UK remains a net importer. Around 4 EUR kg⁻¹, the solution changes regime: European offshore supply rises to about 767 TWh yr⁻¹, outside imports fall to about 63 TWh yr⁻¹, and the UK, Ireland and Denmark re-emerge as large net

exporters, at approximately 162, 161 and 118 TWh yr⁻¹, respectively. The UK/Ireland export conclusion is therefore conditional on external imports not being available at ultra-low delivered prices, but it is restored once the outside option enters a more constrained delivered-cost range.

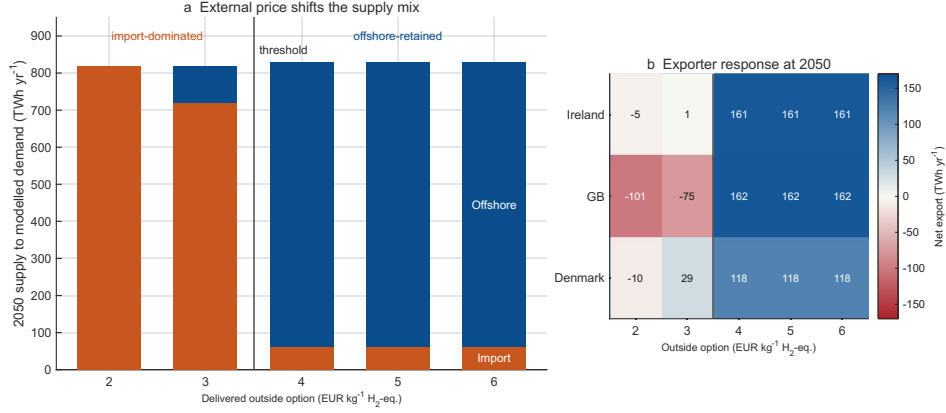


Fig. 8 External import prices define the boundary of European offshore hydrogen supply. **a** 2050 supply mix between external low-carbon imports and European offshore hydrogen under delivered outside-option prices from 2 to 6 EUR kg⁻¹ H₂-equivalent. **b** Net-export response of Ireland, the United Kingdom and Denmark under the same price scan. Blue indicates net export and red indicates net import. The scan identifies a delivered-price threshold rather than assigning probabilities to future import prices.

3 Discussion

The overall analysis shows that offshore green hydrogen can contribute materially to Europe’s net-zero transition, but its strategic geography is not determined by LCOH alone. Denmark remains the strongest cost-competitiveness benchmark across the cost-supply curves, whereas Ireland and the UK become strategically important only after domestic absorption, national demand, shipping and residual external imports are considered together. In the integrated 2050 case, the UK and Ireland become the two largest net exporters with approximately 162 TWh yr⁻¹ each, followed by Denmark with approximately 118 TWh yr⁻¹.

This distinction matters for policy appraisal. A lowest-cost-only allocation would identify Denmark or the Netherlands as the leading exporters and would hide residual import dependence, whereas the integrated model identifies a west-to-east trade structure with persistent external import needs in 2040 and 2050. The result is therefore not that high-resource island systems automatically dominate offshore hydrogen trade. Rather, their strategic role emerges when large offshore build-out coincides with limited domestic absorption, nearby continental demand and a delivered external import option that is not ultra-low cost.

These findings also clarify how Ireland should be interpreted in future hydrogen planning. Ireland has no decisive LCOH advantage and does not have the largest offshore hydrogen production capacity. Its strategic value arises from the combination of

sufficient cost competitiveness, large exportable surplus and European demand centres that remain undersupplied by their domestic offshore resources. This conclusion supports considering multiple export corridors and carrier options rather than evaluating Irish exports only through a single pipeline route. It also implies that national offshore hydrogen strategies should be assessed within a regional trade and demand balance, not only through domestic cost-supply estimates.

The central innovation of this study is therefore methodological as well as quantitative. By linking geospatial offshore hydrogen cost-supply curves, hourly domestic power-gas absorption, hydrogen/ammonia shipping and an explicit external import option, the model exposes a ranking reversal that single-module assessments cannot capture. The lowest-LCOH baseline points to Denmark or the Netherlands as leading exporters, but the integrated model identifies the UK as the leading exporter in all three model years and Ireland as an almost equal exporter by 2050. This first integrated European-scale treatment is what makes the UK/Ireland result visible.

Several assumptions define the scope of these conclusions. The analysis uses currently published offshore wind goals and therefore assumes that planned infrastructure is delivered on schedule. Domestic absorption outside Ireland is estimated by scaling the Ireland hourly whole-system result with national electricity and gas demand, so the export-potential estimates should be read as a European screening result rather than a replacement for country-specific unit-commitment studies. We stress-test this bridge in the Supplementary Information by resolving the integrated trade model after multiplying domestic absorption by 0.75 and 1.25. This changes European offshore supply, residual imports and the exact UK/Ireland ordering, but it does not restore the LCOH-only exporter geography: Ireland remains a leading net exporter and the ranking reversal relative to the cost-only baseline is preserved. The outside-option analysis is also a threshold scan, not a forecast of future global hydrogen prices. These assumptions do not weaken the integrated trade result, but they identify where country-level planning and future market evidence would most improve the analysis.

The results indicate that residual external imports remain necessary in the integrated 2040 and 2050 cases, at approximately 116 and 62 TWh yr⁻¹, respectively. More ambitious offshore hydrogen production goals could reduce this dependence, especially in countries with lower marginal costs and remaining offshore capacity potential, such as Denmark and the Netherlands. Infrastructure policy also matters. Offshore wind turbines, electrolyzers, subsea cables, ports and hydrogen shipping vessels must be coordinated with stable regulation, faster consenting and dedicated export corridors. Otherwise, European hydrogen demand can grow faster than domestic offshore supply.

These findings give a practical test for future hydrogen strategies. Countries should not be ranked only by LCOH or offshore resource quality. They should be evaluated by their position in the wider system: domestic absorption, exportable surplus, carrier choice, shipping access, demand geography and competing external imports. In that system view, Ireland and the UK become strategic exporters because they are system-positioned, not simply cost-ranked.

4 Methods

4.1 Model workflow and scope

We built the workflow to test whether export leadership follows from low offshore production costs alone, or from the interaction between resource quality, domestic absorption, shipping geography and competing imports. The workflow links geospatial supply mapping, hourly domestic system operation, European trade optimisation and carbon-displacement accounting. Offshore wind, bathymetry, port-distance, grid-connection and exclusion-area data define country-level cost-supply curves for offshore hydrogen and ammonia in 2030, 2040 and 2050. A coordinated domestic operation model estimates the share of offshore production that can be absorbed before an export outlet is needed. The remaining export potential enters a European hydrogen and ammonia trade model that also allows demand to be met by an external import option. The optimised delivery matrix is then translated into supplier-attributed carbon-mitigation values. The Methods give the system boundary and assumptions needed to interpret the results; algebraic formulations and parameter tables are reported in the Supplementary Information.

The model covers Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the UK. Each model year is solved independently for 2030, 2040 and 2050. Energy flows are optimised as average power quantities and converted to TWh yr⁻¹ for reporting. Carbon-mitigation outputs are reported in Mt CO₂ yr⁻¹.

4.2 Techno-economic mapping of offshore hydrogen supply

The supply module estimates where offshore hydrogen can be produced and at what cost within the EEZs of the selected coastal European countries [30]. Aquaculture sites, Special Areas of Conservation and military areas are excluded from the developable offshore hydrogen production area using EMODnet boundaries [19]. Within the remaining offshore grid cells, hourly wind velocity, temperature and air-pressure data are taken from the ERA5 reanalysis at 1 h temporal resolution and 0.5° × 0.5° spatial resolution, equivalent to approximately 45 km by 56 km in southern Europe [16].

Hourly wind generation is calculated from the power curve of an NREL reference 15 MW wind turbine [17]. The turbine power curve is defined at 1 m s⁻¹ wind-speed increments, with linear interpolation for intermediate wind speeds. Turbine layouts are chosen by balancing power density against wake losses rather than imposing a uniform turbine spacing. Single-turbine wakes are represented with the Jensen-Gaussian model [31]. For overlapping wakes from multiple upstream turbines, we use a recursive-tree method that reduces simulation time by 79% with less than 1% accuracy loss in benchmark tests [32], avoiding the computational burden of enumerating many wake combinations across long historical weather series [33, 34]. Further methodological details and parameter values are provided in the Supplementary Information.

LCOH is calculated with a bottom-up representation of development expenditure, capital expenditure, operational expenditure and decommissioning expenditure. European techno-economic parameters, including environmental survey, wind-turbine and

cable costs, are taken from Offshore Renewable Energy Catapult [35]; the detailed cost breakdown is provided in the Supplementary Information.

The site-specific cost calculation depends on water depth, capacity factor, distance to the nearest port, distance to the nearest connection point and turbine spacing. Bathymetry data are obtained from EMODnet and converted to $0.001^\circ \times 0.001^\circ$ grid cells, equivalent to approximately 0.11 km by 0.06 km near the UK [18]. Water depth determines whether monopile, jacket or floating turbine structures are technically applicable and affects foundation material requirements. Depths below 30 m are treated as suitable for monopiles, depths of 30–60 m for jacket foundations, and depths above 60 m for floating foundations. Port distance affects installation and maintenance costs, while distance to connection points affects export-cable CAPEX. Turbine spacing is selected so that wake interference from adjacent turbines remains below 5%, which also determines the array-cable length used in the cost calculation.

Offshore hydrogen production is represented as an integrated wind-electrolysis platform with water pumping and desalination, alkaline electrolyzers, hydrogen compressors, standby batteries and auxiliary offshore infrastructure. Alkaline electrolysis is used because it is currently a mature and durable electrolyser option for large-scale deployment; techno-economic characteristics are summarised in Supplementary Table 4. Capital-cost and electrical-efficiency projections are taken from Offshore Renewable Energy Catapult and the UK’s National Hydrogen Strategy, with alkaline-electrolyser efficiency increasing from 0.79 in 2030 to 0.82 in 2050 [36]. Annual OPEX includes maintenance, stack replacement and labour costs as a percentage of initial capital investment.

Because variable offshore wind output can differ from the electrolyser operating point, the platform includes electrical and hydrogen storage. Water pumping, desalination and compression loads are sized with hydrogen production, so their power consumption scales with hydrogen production capacity. Algebraic sizing equations are used to maximise utilisation and determine the electrolyser and accessory capacities (Supplementary Information, Section 3.3). Future operating parameters and cost projections are taken from Offshore Renewable Energy Catapult [35]. LCOH is then calculated for the combined wind farm, electrolysis system and offshore auxiliaries.

Cost-supply curves are generated by ranking grid-cell production segments by LCOH and accumulating the available hydrogen quantity. The potential installed offshore wind capacity for each grid cell is calculated from turbine rated power and the maximum number of turbines that can be placed in the cell under the wake-interference constraint. Vessel density, obtained from EMODnet at 1 km resolution, reduces available power density linearly up to a maximum 50% reduction [37, 38]. Future LCOH reductions are represented through high, median and low technology-learning scenarios based on expert elicitation [21]. Under the median scenario, relative to 2020, fixed-bottom offshore wind costs are reduced by 21%, 35% and 49% in 2030, 2040 and 2050, respectively; floating offshore wind costs are reduced by 2%, 17% and 40%.

4.3 Domestic absorption of offshore wind and hydrogen

The domestic model quantifies a binding system limit: how much offshore energy can be used inside the All-Island power and gas systems before exports create additional

value. Offshore electricity is connected to the power system by marine cables, while offshore hydrogen can enter the gas system through blending or, in the 2050 case, a pure-hydrogen network assumption. The scenario hydrogen fraction in the gas network is set to 0% in 2030, 20% in 2040 and 100% in 2050. These cases represent a transition from no material blending, through an intermediate blending limit, to a dedicated hydrogen-network case. The 20% case reflects current blending demonstrations and policy discussion [39–42], while the 2050 case is aligned with hydrogen-network planning [43].

The coordinated model determines hourly operation over a representative year while minimising operating cost and maximising offshore-resource integration subject to power-system and gas-system constraints. We use paired export and no-export cases to separate offshore wind that can be absorbed domestically from offshore wind that requires an external hydrogen or ammonia outlet. The resulting quantities define domestic offshore wind use, hydrogen production, curtailment and exportable surplus.

The power side is represented by a nodal unit-commitment model that balances electricity demand and uncertain renewable generation. The objective includes thermal-unit fuel costs and start-up costs, with operational constraints for ramping, minimum online and offline times, minimum output levels, must-run units and reserves. The model also represents the operational policy roadmap for the All-Island system, including inertia, minimum numbers of conventional units and System Non-Synchronous Penetration limits [44]. Network congestion is represented with DC power flow. Seasonal dynamic line ratings are included for summer and winter cases, and planned interconnection expansion, including the 700 MW Celtic Interconnector with France, is represented [45].

The gas side captures the operational effect of hydrogen injection from offshore electrolysis. Hydrogen blending changes gas composition and therefore changes the heat value, specific gravity and safety characteristics of the gas mixture. We define an admissible gas-quality region using the Gas Safety (Management) (Amendment) Regulations 2023, including relative density, Wobbe index and Weaver flame-speed factor limits [46]. Gas flow and pressure are represented with the Weymouth equation. Because variable gas composition introduces additional non-convexity, the gas-flow problem is solved using the previously developed reformulation framework based on McCormick envelopes, second-order cone reformulation and sequential convex programming [47]. Gas streams with different compositions are mixed at pipeline junctions; the resulting bilinear terms are handled with the same solution framework. Mathematical details are provided in Supplementary Information, Section 4.

The coordinated operation model is parameterised with detailed All-Island generation, demand and network data. Generator capacities, piecewise heat-rate curves and operating parameters are obtained from the Single Electricity Market model used in EirGrid publications [48]. Commodity costs and heat values are represented for non-gas fossil-fuel units. For gas-fired units, fuel use is linked to the gas-system calculation so that heat value changes with hydrogen fraction. Onshore wind and solar generation are calculated from ERA5 wind speed and solar-radiation data; photovoltaic output is calculated using the Beckman model. Hydropower generation is taken from historical operation data [49].

Future gas demand is based on Gas Networks Ireland’s Gas Forecast Statement and reflects coal-plant retirements, the 80% renewable-generation ambition in Ireland’s Climate Action Plan, restrictions on new gas boiler connections, GDP-linked residential, industrial and commercial demand changes, and compressed natural gas station development [50]. Daily electricity peak demands to 2032 are taken from EirGrid’s Ten-Year Generation Capacity Statement for winter, summer and spring-autumn cases [51]. Missing values between 2033 and 2050 are filled using an autoregressive integrated moving average model. Fifteen-minute electricity and gas demand profiles are taken from EirGrid’s smart-grid dashboard and the Gas Networks Ireland data dashboard [52, 53]. High-voltage power lines above 110 kV and high-pressure gas pipelines above 16 bar are represented using network topology, location and parameter data from EirGrid and Gas Networks Ireland [54, 55]. Gas-pipeline friction factors and diameters follow an Ireland case study [56]; power-line capacities are assumed to grow at the same annual rate as demand.

4.4 European hydrogen and ammonia trade optimisation

The trade model asks where residual offshore supply is used once domestic absorption has been accounted for. It combines the country-level cost-supply curves and export-potential estimates with hydrogen and ammonia demand, shipping costs, domestic self-supply and an external import option in 2030, 2040 and 2050. Countries with published offshore wind targets, including Ireland [57] and the UK [58], use those targets as maximum offshore capacities. For countries without complete year-specific targets, we use tentative goals from the European Union’s non-binding agreement on offshore wind deployment [59]. Hydrogen and ammonia demands are taken from the European Hydrogen Backbone demand projection [60].

The optimisation minimises total production, shipping and external-import costs subject to national offshore supply capacities, domestic export limits, country-level demand, shipping-route capacities and material balances. National production, bilateral carrier flows, route-level vessel deployment and external imports are chosen simultaneously. Hydrogen and ammonia production costs are represented through piecewise-linear interpolations of the cost-supply curves. Shipping costs include annualised vessel investment costs and operating fuel costs, with route distances derived from port-to-port shipping paths [19]. Because the supply curves and vessel-count decisions introduce discrete choices, the model is formulated as a mixed-integer linear programming problem and solved in MATLAB using Gurobi. The complete formulation is given in Supplementary Information, Section 5.

The international import node acts as an outside option rather than as an additional European supplier. It is retained in the base integrated model so that demand can still be met when European offshore supply is insufficient. In the outside-option sensitivity, the same node is assigned a delivered hydrogen-equivalent price and the model is solved repeatedly over 2–6 EUR kg^{−1} H₂-equivalent, corresponding to 60–180 EUR MWh^{−1}. Hydrogen and ammonia imports are assigned the same hydrogen-equivalent delivered price so that the scan isolates the threshold at which European offshore supply becomes competitive with non-European low-carbon imports.

We use carbon mitigation as an attribution metric rather than a full lifecycle assessment. Optimised hydrogen and ammonia deliveries are converted to avoided CO₂ using the carrier-specific conversion factors defined in the Supplementary Information and assigned to a supplier-destination contribution matrix. Positive bilateral route variables assign the avoided CO₂ to the exporting country and importing country; negative route variables reverse the direction before assignment. Domestic self-supply is assigned to the matrix diagonal, and the external import node is reported separately from the 11 European offshore supplier countries. The total carbon mitigation reported in the main text is the sum of the full matrix, while the European offshore supplier contribution excludes the external import row.

To separate the integrated result from a cost-only interpretation, we construct a lowest-LCOH counterfactual in which total European hydrogen and ammonia demand is allocated to offshore supply segments in ascending LCOH order. This baseline uses planned offshore capacity and capacity factors to limit country supply, but deliberately excludes domestic absorption limits, shipping costs, demand geography and the competitive external-import price. It is therefore used only as a diagnostic baseline for testing whether exporter rankings can be inferred from production cost alone.

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Author contributions

S.W. conceived the study, developed the modelling framework and MATLAB code, performed the analysis, prepared the figures and wrote the original manuscript. M.M.B. contributed to the energy-system interpretation, policy context and manuscript review and editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The public data sources used in this study are cited in the References and described in the Methods and Supplementary Information. Derived source-data tables generated by the analysis, including the headline-number audit table used to trace manuscript values and a source-data coverage inventory, are available in the project repository under **results/tables/**. Final figure files are provided under **manuscript/figs/**. Large public raw climate, bathymetry and geospatial inputs and large intermediate MATLAB checkpoint files are not redistributed through the GitHub repository because of file-size constraints; the raw inputs are available from the cited public data providers and the checkpoints can be regenerated from the provided code.

Code availability

MATLAB code used in this study is available on GitHub at <https://github.com/ShengWang-EE/HyExport>.

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